
InvenTrack: Inventory Management System Using DAO Pattern with File Persistence

Course: PCCCS495: Advanced Programming (OOP) Lab

Term II Project

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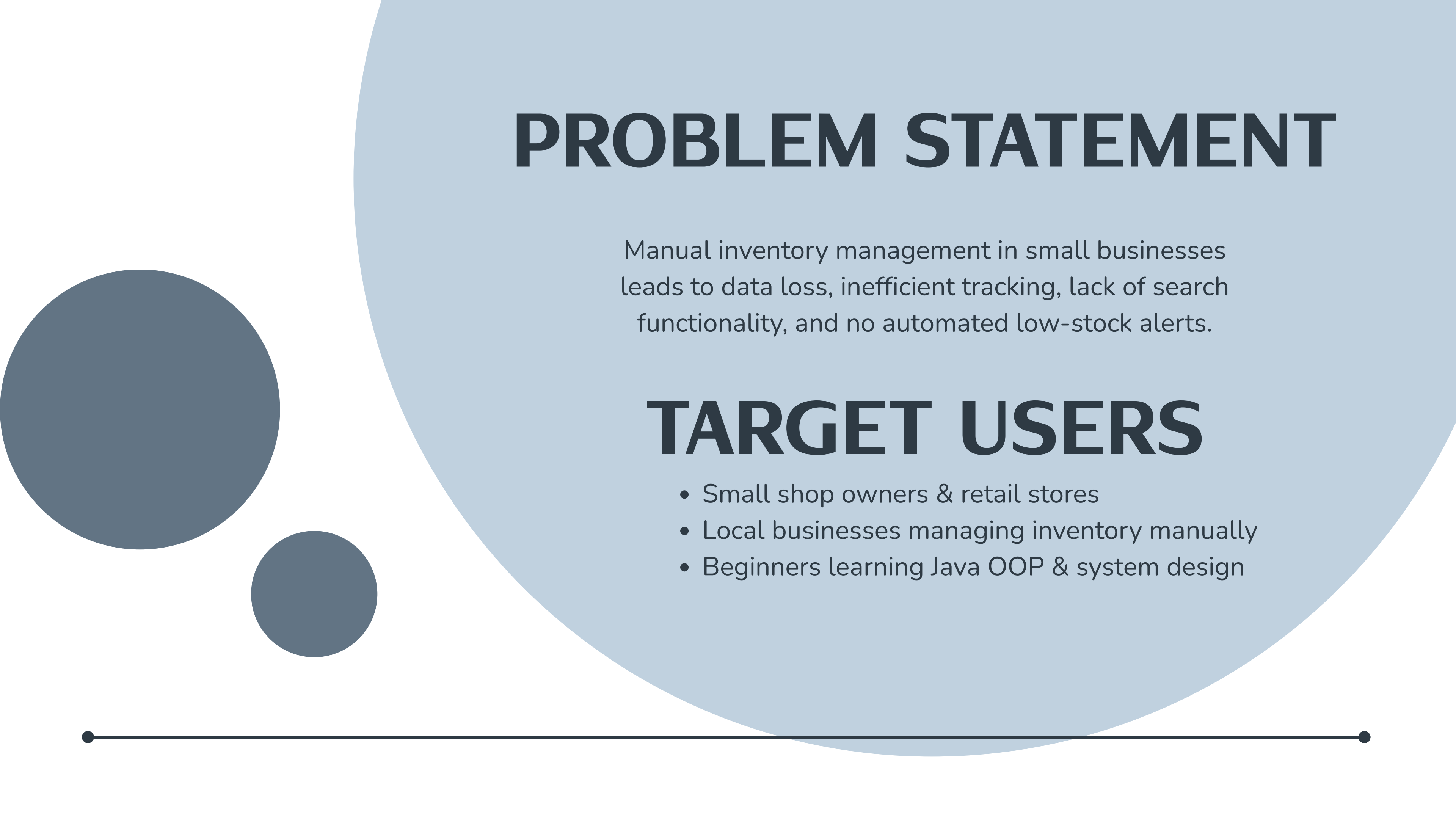
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PROBLEM STATEMENT

Manual inventory management in small businesses leads to data loss, inefficient tracking, lack of search functionality, and no automated low-stock alerts.

TARGET USERS

- Small shop owners & retail stores
 - Local businesses managing inventory manually
 - Beginners learning Java OOP & system design
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SYSTEM ARCHITECTURE

Model Layer:

Responsibility: Data representation using Product, Category, Supplier classes.

DAO Layer:

Responsibility: Data access and persistence through interfaces and file persistence.

Service Layer:

Responsibility: Contains core business logic and validation rules.

UI Layer:

Responsibility: Presents information and handles user interaction via a Swing-based dashboard.



OOP CONCEPTS DEMONSTRATED



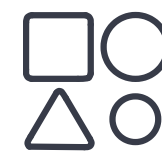
Abstraction

DAO interfaces declare methods without implementation.



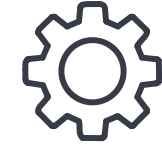
Inheritance

ValidationException extends RuntimeException, UI panels extend JPanel.



Polymorphism

Interface references resolved at runtime, method overriding in UI panels.



Design Pattern (DAO)

DAO pattern for data access abstraction.

TECHNICAL HIGHLIGHT

Generic FileManager (Java Generics)

- Single reusable method for all entities (Product, Category, Supplier)
- Reduces code duplication and improves maintainability

Service Layer Validation

- Centralized validation for all inputs
- Ensures data integrity & prevents invalid operations
- Enforces referential integrity (valid category & supplier)

Unique ID Generation

- Auto-generated sequential IDs (PRD001, PRD002...)
- Prevents duplication even after deletions
- Ensures consistent product tracking

Low Stock Detection Logic

- Encapsulated inside Product class
- Reusable across Dashboard & Low Stock panel
- Provides real-time inventory alerts

LIVE DEMO

InvenTrack

Dashboard

Products

Categories

Suppliers

Low Stock

Suppliers

Add Supplier

ID	Company Name	Contact Person	Phone	Email
SUP001	Samsung India	Rajesh Kumar	9876543210	rajesh@samsung.in
SUP002	Fresh Farms Co.	Priya Sharma	9123456780	priya@freshfarms.com
SUP003	Trends	Arya Kumar	1234567890	arya123@gmail.com
SUP004	CocaCola		7896143670	

Edit

Delete

InvenTrack

Dashboard

Products

Categories

Suppliers

Low Stock

Dashboard

Total Products

5

Categories

4

Suppliers

4

Low Stock Alerts

2

Total Inventory Value: ₹4,193,945.00

Products Needing Restock

Product ID	Product Name	Current Stock	Min. Required
PRD003	USB-C Chargin...	5	10
PRD005	Thumbs Up	0	5

InvenTrack

Dashboard

Products

Categories

Suppliers

Low Stock

Products

Add Product

Search:

Search

Category: All Categories

Reset

ID	Name	Category	Supplier	Stock	Price (₹)	Status
PRD001	Samsung Galaxy S24	Electronics	Samsung India	50	79999.00	OK
PRD002	Basmati Rice 5kg	Groceries	Fresh Farms Co.	200	450.00	OK
PRD003	USB-C Charging Cable	Electronics	Samsung India	5	599.00	Low
PRD004	Jacket	Clothing	Trends	101	1000.00	OK
PRD005	Thumbs Up	Cold Drinks	CocaCola	0	200.00	Low

Edit

Delete

Restock

Export CSV

InvenTrack

Dashboard

Products

Categories

Suppliers

Low Stock

Low Stock Alerts

Refresh

2 product(s) need restocking immediately.

Products below are at or below their minimum stock threshold. Click a row and press 'Restock' to update the quantity.

Product ID	Product Name	Category	Supplier	Current Stock	Min. Required	Shortage
PRD003	USB-C Charging Cable	Electronics	Samsung India	5	10	-5 units
PRD005	Thumbs Up	Cold Drinks	CocaCola	0	5	-5 units

Restock Selected

-- Select a product row first

InvenTrack

Dashboard

Products

Categories

Suppliers

Low Stock

Categories

Add Category

ID	Name	Description
CAT001	Electronics	Electronic gadgets and devices
CAT002	Groceries	Food and daily essentials
CAT003	Clothing	Good clothes
CAT004	Cold Drinks	Refreshments

Edit

Delete



**Thank
You**